Plataformas de Desenvolvimento

25/26

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Mestrado em Informática e Sistemas

1

General Objectives

- Understanding the concepts of Continuous Integration / Continuous Delivery (CI/CD)
- Implementing a CI/CD pipeline
- Understanding concepts about Cloud Computing
  - Specifically AWS
- Create an application based on AWS services

2

General Objectives

- General Topics:
  - Docker (concepts; creation; use of Docker images)
  - Ansible (concepts; specifying macros for deployment in environments)
  - Jenkins (concepts and practical exercises)
  - Git (concepts and practical exercises)
  - Github Actions (concepts and practical exercises)
  - Netlify (concepts and practical exercises)
  - Cloud computing notions
  - AWS Cloud services (Database, Lambda, API Gateway, etc.)

Highly focused on technologies and experimentation.

3

Evaluation

- Written / in-person exam - 7 points
- Practical assignments - 13 points

4

Evaluation

- Written / in-person exam - 7 points
  - Minimum of 25%

5

Evaluation

- Practical Assignments - 13 points
  - Includes:
    - CI/CD development component (6 points)
    - AWS development component (5 points)
    - Mini-laboratories (1 point)
    - GitHub Actions (1 points)

    - Development components are done in groups, but grades may vary within the group.
    - Mini-laboratories are done individually.

  - Presentation schedule

Component	Presentation	Report and other artefacts
CI/CD	16/04 or 21/05	7/06
AWS	21/05 or 04/06	7/06
Mini-labs	No presentation	21/05
GitHub Actions	21/05	No Report

6

Evalution

- The practical assignment involves:

- CI/CD development component
  - Proposal, planning, and development of a CI/CD pipeline
  - A BackEnd, FrontEnd, and Database application.
  - In addition to implementation, the report and presentation are crucial.

7

Evaluation

- AWS component

  - Create an application using cloud services.

  - In addition to implementation, the report and presentation are crucial.

8

Classes

- The classes have a strong practical aspect.
- The initial part of the class will be more explanatory, with the use of document projection and examples.
- The final part of the class will be more practical, focusing on solving exercises.
- Some days will be entirely dedicated to exercises.
- The last classes of each module will be reserved for presenting the research component and finalizing the practical assignment.
- Classes should end before 10:25 PM.

9

Workbench Environment

Linux Virtual Machine environment to setup several tools.

- On each student' laptop

- Linux 64 bits.

10

First Class

- VM Installation and setup

11

Bibliography

- Docker in Practice, 2nd Ed., Ian Miell, Aidan Hbson Sayers, Manning 2019
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- https://www.oreilly.com/library/view/building-microservices-2nd/9781492034018/
- https://aws.amazon.com/

12